

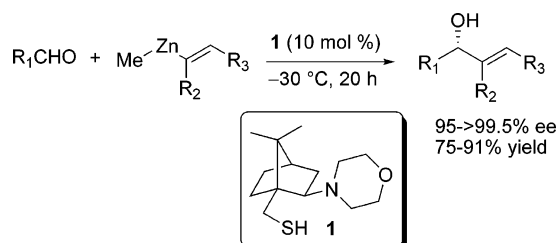
Highly Enantioselective Synthesis of (*E*)-Allylic Alcohols[†]

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The asymmetric addition of alkenylzincs to aromatic and α -branched aliphatic aldehydes catalyzed by **1** generated the corresponding (*E*)-allylic alcohols with >95% ee and good to excellent chemical yields, especially >99.5% ee was observed in the case of 4-CF₃-benzaldehyde. Notably, **1** is an effective ligand to catalyze the addition of disubstituted ($R_2 = R_3 =$ ethyl) and bulky substituted ($R_2 =$ H, $R_3 =$ *tert*-butyl) alkenylzincs to benzaldehyde, affording the corresponding allylic alcohols both with 96% ee.

Since the discovery of Et₂Zn,¹ the preparation² of organozinc reagents and their applications³ in C–C bond formation have been well-investigated. Unlike the enantioselective addition of alkyllithium and Grignard reagents, addition of alkylzincs to aldehydes in the presence of chiral catalysts⁴ affords chiral secondary alcohols, possessing diversity of functionality, in high enantioselectivities. Catalytic asymmetric addition of organozinc reagents to carbonyl compounds has been widely studied for over two decades, and successful examples of catalysts with various ligand scaffolds have been reported.⁴ Recently, the enantioselective addition of alkenylzinc reagents to aldehydes

has drawn the attention of organic chemists. The advantages of this methodology include the following: (1) the alkenylzinc reagents could be prepared in situ with simple operations either by metal exchange of zinc chloride with alkenyl magnesium bromide, zinc bromide with alkenyllithium,⁵ or by reaction of dialkylzinc with alkenylzirconium reagents⁶ or alkenylboranes;⁷ and (2) a regioselective synthesis of *E*-double bonds as well as an asymmetric carbon–carbon bond formation occurred during the course of reaction, and chiral allylic alcohols thus obtained are important intermediates for the preparation of numerous natural products and biologically active compounds.⁸ Employment of various categories of ligands such as β -amino alcohols,^{9,10} paracyclophane-based ketimines,¹¹ β -amino thiols,¹² γ -amino naphthols,¹³ and dipeptide ligands^{14,15} for such purposes have been studied. However, the exploration of efficient chiral ligands to synthesize allylic alcohols in good enantioselectivities still remains a challenge that requires considerable and intensive investigation. Additionally, it has been demonstrated that further

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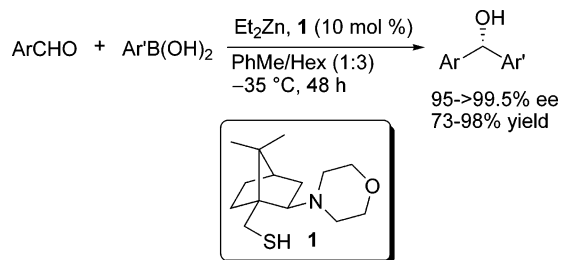
[†] In memory of Professor Yoshihiko Ito who passed away on December 23, 2006.

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SCHEME 1. Asymmetric Arylation of Aldehydes Catalyzed by **1**

manipulations of the chiral allylic alcohols to synthetically useful and valuable enantioenriched α -amino acids,^{10b} γ -unsaturated β -amino acids,^{10d} epoxy alcohols,^{10e,f} and β -hydroxyaldehydes^{10g} could be achieved, and these methods represent advances in terms of synthetic efficiency and selectivity.

As part of our efforts on the development and scrutiny of camphor-derived chiral ligands for catalytic asymmetric reactions,¹⁶ (–)-2-*exo*-morpholinoisoborne-10-thiol (**1**) was found to be an effective chiral ligand in the arylation of aldehydes with up to >99.5% ee (Scheme 1).¹⁷ To study the scope and catalytic ability of **1**, we investigate the application of **1** in alkenylation reaction, and herein, we describe the optimization of reaction conditions and scope of **1** in the alkenylation of aldehydes.

The in situ preparation of reactive alkenylzinc species starts from hydroboration of 1-octyne with dicyclohexylborane,^{7a} followed by boron–zinc exchange reaction with Et₂Zn to yield active ethyl 1-octenylzinc. At the outset, reaction of 1-octenylzinc with benzaldehyde catalyzed by 10 mol % of **1** in hexane was examined at –30 °C, and (*R,E*)-1-phenyl-2-nonene-1-ol (**2a**) was obtained with 91% yield and 91% ee (Table 1, entry 1). Allylic alcohol **2a** was obtained with slightly better ee when the reaction was conducted in toluene (entry 2). To examine the influence of dimethyl sulfide (DMS), from BH₃·DMS complex in the preparation of dicyclohexylborane, DMS was removed after preparation of 1-octenylborane, and with that, a slight improvement of the enantioselectivity was observed (entry 3). Lowering reaction temperature did not improve enantioselectivity, and decreasing catalytic loading of **1** diminished the enantioinduction (entries 4–6). Finally, transmetalation using dimethylzinc provided a better selectivity than using diethylzinc (entry 7).

In light of the above observations, addition of 1-octenylzinc to various aldehydes was examined. As shown in Table 2, addition of 1-octenylzinc reagent to substituted benzaldehydes with either electron-donating or electron-withdrawing groups gave the corresponding allylic alcohols **2a–2j** with high enantioselectivities and yields (entries 1–10), while catalytic addition of methyl 1-hexenylzinc to benzaldehyde and 3-CF₃-benzaldehyde provided **2k** and **2l** with 97 and 98% ee, respectively (entries 11 and 12). Notably, in the case of 4-CF₃-benzaldehyde, >99.5% ee of the allylic alcohol **2j** was observed (entry 10).

TABLE 1. Optimization of the Reaction for the Asymmetric Alkenylation of Benzaldehyde Catalyzed by **1**

entry	solvent	temp (°C)	yield (%) ^a	ee (%) ^b
1	hexane	–30	91	91
2	toluene	–30	88	93
3 ^c	toluene	–30	86	94
4 ^c	toluene	–20	88	89
5 ^c	toluene	–40	70	90
6 ^{c,d}	toluene	–30	81	90
7 ^{c,e}	toluene	–30	85	96

^a Isolated yield. ^b Determined by chiral HPLC; see Supporting Information. ^c DMS was removed following the preparation of alkenylborane. ^d 5 mol % of **1** was used. ^e Me₂Zn was used instead of Et₂Zn.

TABLE 2. Asymmetric Alkenylation of Various Aldehydes

entry	R ₁	R ₂	R ₃	allylic alcohol	yield (%) ^a	ee (%) ^b
1	H	C ₆ H ₁₃	Ph	2a	85	96
2	H	C ₆ H ₁₃	2-Cl–Ph	2b	85	95
3	H	C ₆ H ₁₃	3-Cl–Ph	2c	88	97
4	H	C ₆ H ₁₃	4-Cl–Ph	2d	90	96
5	H	C ₆ H ₁₃	3-F–Ph	2e	88	96
6	H	C ₆ H ₁₃	2-Me–Ph	2f	83	96
7	H	C ₆ H ₁₃	4-Me–Ph	2g	83	95
8	H	C ₆ H ₁₃	3-MeO–Ph	2h	81	95
9	H	C ₆ H ₁₃	3-CF ₃ –Ph	2i	87	98
10	H	C ₆ H ₁₃	4-CF ₃ –Ph	2j	90	>99.5
11	H	C ₄ H ₉	Ph	2k	83	97
12	H	C ₄ H ₉	3-CF ₃ –Ph	2l	82	98
13	C ₂ H ₅	C ₂ H ₅	Ph	2m	89	96
14	H	^t Bu	Ph	2n	91	96
15	H	Ph(CH ₂) ₃	cyclohexyl	2o	80	97

^a Isolated yield. ^b Determined by chiral HPLC; see Supporting Information.

To further study the scope of the reaction, we investigated addition of bulky substituted and disubstituted alkenylzincs to aromatic and α -branched aliphatic aldehydes. Excellent enantioselectivities and chemical yields were also observed in the cases of addition of benzaldehyde with alkenylzincs that were prepared from *tert*-butylethyne, a bulky alkyne, or 3-hexyne, an internal alkyne (entries 13 and 14). Catalytic asymmetric alkenylation of cyclohexane carboxaldehyde yielded **2o** with 97% ee¹⁸ (entry 15). To the best of our knowledge, **1** is one of the effective and efficient ligands to achieve enantioselective alkenylation of both bulky and internal alkenylzincs with benzaldehyde.^{6e,10b,11b,12,14b,c}

Additionally, *ent*-**1** was easily prepared from (1*R*)-(–)-10-camphorsulfonic acid¹⁷ and that provides access to both (*R*) and (*S*)-chiral allylic alcohol.

In conclusion, we have achieved a highly enantioselective alkenylation of aromatic and α -branched aliphatic aldehydes

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catalyzed by **1** with alkenylzincs that were prepared either from internal or terminal alkynes. Our method produced chiral allylic alcohols, bearing *R*-configuration,¹⁸ in 95 to >99.5% ee¹⁹ and 80–92% yield. This finding offers a useful synthetic route to highly functionalized (*E*)-chiral allylic alcohols.

Experimental Section

A representative example for the alkenylation of aldehydes was conducted as follows: cyclohexene (300 μ L, 3.0 mmol) was slowly added to a flask that contained hexanes (1.0 mL) and BH₃·DMS (1.5 mmol) at 0 °C. After being stirred at 0 °C for 3 h, 1-octyne (220 μ L, 1.5 mmol) was added to the flask, and the mixture was slowly warmed to room temperature. The mixture was stirred at room temperature for 1 h, and the volatile was evaporated in vacuo. Toluene (1.0 mL) was added to the residue, and the resulting mixture was cooled to –78 °C. Dimethylzinc solution (1.5 mmol, 2.0 M solution in toluene) was added to the above mixture followed by a solution of **1** (25 mg, 0.098 mmol in 0.5 mL of toluene). The mixture was warmed to –30 °C over a period of 1 h, and benzaldehyde (100 μ L, 0.98 mmol) in toluene (4.0 mL) was added

to the reaction mixture for 1 h using a syringe pump. After being stirred at –30 °C for 20 h, the reaction was quenched by adding saturated ammonium chloride (5 mL). The aqueous solution was extracted with ether (3 $\times$ 10 mL). The combined ether extracts were dried and concentrated in vacuo. The crude product was purified by column chromatography (EtOAc/hexanes = 1:8) to yield (*R,E*)-1-phenylnon-2-en-1-ol **2a** (182 mg, 85%) as a colorless oil. The enantioselectivity was measured by HPLC (Chiralcel OB, 2-propanol/hexanes = 1:99, 1.0 mL/min) to be 96% ee: [α]_D²⁴ –32.8 (c 1.9, CHCl₃); ¹H NMR (500 MHz, CDCl₃) δ = 7.48–7.14 (m, 5H), 5.75 (dt, *J* = 14.7, 6.5 Hz, 1H), 5.65 (ddd, *J* = 14.7, 6.5, 1.5 Hz, 1H), 5.14 (d, *J* = 6.5 Hz, 1H), 2.04 (dt, *J* = 6.5, 6.5 Hz, 2H), 1.46–1.14 (m, 9H), 0.87 (t, *J* = 6.0 Hz, 3H); ¹³C NMR (125 MHz, CDCl₃) δ = 143.4 (C), 132.8 (CH), 132.2 (CH), 128.5 (CH), 127.4 (CH), 126.1 (CH), 75.2 (CH), 32.2 (CH₂), 31.6 (CH₂), 28.9 (CH₂), 28.8 (CH₂), 22.6 (CH₂), 14.0 (CH₃); IR (neat) 3362, 3085, 3062, 3029, 2956, 2926, 1668, 1492 cm^{–1}; HRMS calcd for C₁₅H₂₂O 218.1671, found 218.1665.

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Supporting Information Available: Experimental details and full characterizations of new compounds. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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